

Jupyter Notebook Execution Report

Name: Your Name

Date: March 31, 2026

Cell 1: ■ Code

```
import pandas as pd

#
# pd.set_option("display.max_rows", None)
# pd.set_option("display.max_columns", None)
# pd.set_option("display.width", None)
# pd.set_option("display.max_colwidth", None)

DEMO_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/DEMO_L.xpt", format="xport")

DIQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/DIQ_L.xpt", format="xport")

GHB_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/GHB_L.xpt", format="xport")

GLU_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/GLU_L.xpt", format="xport")

HIQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/HIQ_L.xpt", format="xport")

HUQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/HUQ_L.xpt", format="xport")

INQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/INQ_L.xpt", format="xport")

UCPREG_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/UCPREG_L.xpt", format="xport")

BMX_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/BMX_L.xpt", format="xport")

PAQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/PAQ_L.xpt", format="xport")

WHQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/WHQ_L.xpt", format="xport")
```

```
SLQ_21_23 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/SLQ_L.xpt", format="xport")

DEMO_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/DEMO_J1718.xpt", format="xport")

DIQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/DIQ_J1718.xpt", format="xport")

GHB_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/GHB_J1718.xpt", format="xport")

GLU_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/GLU_J1718.xpt", format="xport")

HIQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/HIQ_J1718.xpt", format="xport")

HUQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/HUQ_J1718.xpt", format="xport")

INQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/INQ_J1718.xpt", format="xport")

UCPREG_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/UCPREG_J1718.xpt", format="xport")

BMX_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/BMX_J1718.xpt", format="xport")

PAQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/PAQ_J1718.xpt", format="xport")

WHQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/WHQ_J1718.xpt", format="xport")

SLQ_17_18 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/SLQ_J1718.xpt", format="xport")

P_WHQ_17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_WHQ17_20.xpt", format="xport")

P_SLQ_17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_SLQ17_20.xpt", format="xport")

P_PAQ17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_PAQ17_20.xpt", format="xport")

P_INQ17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_INQ17_20.xpt", format="xport")

P_HIQ17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_HIQ17_20.xpt", format="xport")

P_GLU17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_GLU17_20.xpt", format="xport")

P_GHB17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_GHB17_20.xpt", format="xport")
```

```

P_DIQ17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_DIQ17_20.xpt", format="xport")

P_DEM017_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_DEM017_20.xpt", format="xport")

P_UCPREG17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_UCPREG17_20.xpt", format="xport")

P_INS17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_INS17_20.xpt", format="xport")

P_BMX17_20 = pd.read_sas("/Users/siddarthusulkar/Documents/Project files/Data
Sets/P_BMX17_20.xpt", format="xport")

```

Error:

Traceback (most recent call last):

```

File "/Users/unnatiagarwal/.vscode/extensions/ganeshkumbhar.nb2pdf-1.1.9/scripts/nb2pdf.py", li
exec('\n'.join(lines[:-1]), glb)
File "<string>", line 8, in <module>;
File "/Users/unnatiagarwal/Documents/Semester_4/Applied_Project/Untitled/.venv/lib/python3.12/s
reader = XportReader(
    ^^^^^^^^^^^^^^^
File "/Users/unnatiagarwal/Documents/Semester_4/Applied_Project/Untitled/.venv/lib/python3.12/s
self.handles = get_handle(
    ^^^^^^^^^^^^^^^
File "/Users/unnatiagarwal/Documents/Semester_4/Applied_Project/Untitled/.venv/lib/python3.12/s
handle = open(handle, ioargs.mode)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

```

FileNotFoundError: [Errno 2] No such file or directory: '/Users/siddarthusulkar/Documents/Project

Cell 2: ■ Code

```

df = DEMO_21_23

df = df.merge(DIQ_21_23, on="SEQN", how="left")
df = df.merge(GHB_21_23, on="SEQN", how="left")
df = df.merge(GLU_21_23, on="SEQN", how="left")
df = df.merge(HIQ_21_23, on="SEQN", how="left")
df = df.merge(HUQ_21_23, on="SEQN", how="left")
df = df.merge(INQ_21_23, on="SEQN", how="left")
df = df.merge(UCPREG_21_23, on="SEQN", how="left")
df = df.merge(BMX_21_23, on="SEQN", how="left")
df = df.merge(PAQ_21_23, on="SEQN", how="left")
df = df.merge(WHQ_21_23, on="SEQN", how="left")
df = df.merge(SLQ_21_23, on="SEQN", how="left")

```

```
df.shape
```

Error:

Traceback (most recent call last):

File "/Users/unnatiagarwal/.vscode/extensions/ganeshkumbhar.nb2pdf-1.1.9/scripts/nb2pdf.py", line 1, in
exec('\n'.join(lines[:-1]), glb)

File "<string>", line 1, in <module>

NameError: name 'DEMO_21_23' is not defined

Cell 3: ■ Code

```
df2 = DEMO_17_18  
df2 = df2.merge(DIQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(GHB_17_18, on="SEQN", how="left")  
df2 = df2.merge(GLU_17_18, on="SEQN", how="left")  
df2 = df2.merge(HIQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(HUQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(INQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(UCPREG_17_18, on="SEQN", how="left")  
df2 = df2.merge(BMX_17_18, on="SEQN", how="left")  
df2 = df2.merge(PAQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(WHQ_17_18, on="SEQN", how="left")  
df2 = df2.merge(SLQ_17_18, on="SEQN", how="left")  
  
print(f"17_18 merged dataset shape: {df2.shape}")
```

Error:

Traceback (most recent call last):

File "/Users/unnatiagarwal/.vscode/extensions/ganeshkumbhar.nb2pdf-1.1.9/scripts/nb2pdf.py", line 1, in
exec('\n'.join(lines[:-1]), glb)

File "<string>", line 1, in <module>

NameError: name 'DEMO_17_18' is not defined

Cell 4: ■ Code

```
df3 = P_DEM017_20  
df3 = df3.merge(P_DIQ17_20, on="SEQN", how="left")  
df3 = df3.merge(P_GHB17_20, on="SEQN", how="left")  
df3 = df3.merge(P_GLU17_20, on="SEQN", how="left")  
df3 = df3.merge(P_HIQ17_20, on="SEQN", how="left")
```

```

df3 = df3.merge(P_INQ17_20, on="SEQN", how="left")
df3 = df3.merge(P_UCPREG17_20, on="SEQN", how="left")
df3 = df3.merge(P_INS17_20, on="SEQN", how="left")
df3 = df3.merge(P_BMX17_20, on="SEQN", how="left")
df3 = df3.merge(P_PAQ17_20, on="SEQN", how="left")
df3 = df3.merge(P_SLQ_17_20, on="SEQN", how="left")
df3 = df3.merge(P_WHQ_17_20, on="SEQN", how="left")

print(f"17_20 (P_*) merged dataset shape: {df3.shape}")

```

Error:

```

Traceback (most recent call last):
  File "/Users/unnatiagarwal/.vscode/extensions/ganeshkumbhar.nb2pdf-1.1.9/scripts/nb2pdf.py", line 1, in <module>
    exec('\n'.join(lines[:-1]), glb)
  File "<string>", line 1, in <module>
NameError: name 'P_DEMO17_20' is not defined

```

Cell 5: ■ Code

```

df3.to_csv("Cleaning_data_merged_17_20.csv", index=False)
print("Saved cleaned data to Cleaning_data_merged_17_20.csv")

```

Error:

```

Traceback (most recent call last):
  File "/Users/unnatiagarwal/.vscode/extensions/ganeshkumbhar.nb2pdf-1.1.9/scripts/nb2pdf.py", line 1, in <module>
    exec('\n'.join(lines[:-1]), glb)
  File "<string>", line 1, in <module>
NameError: name 'df3' is not defined

```

Cell 6: ■ Code

```

import pandas as pd

# Load datasets
df1 = pd.read_csv("Cleaning_data_merged.csv")
df2 = pd.read_csv("Cleaning_data_merged_17_20.csv")

# Check shapes
print("Shape of first dataset:", df1.shape)
print("Shape of second dataset:", df2.shape)

# Merge by stacking rows
combined_df = pd.concat([df1, df2], axis=0, ignore_index=True)

```

```
# Check final shape
print("Shape after merging:", combined_df.shape)

# Save merged file
combined_df.to_csv("Cleaning_data_17_covid_23.csv", index=False)

print("Merged file saved as Cleaning_data_17_covid_23.csv")
```

Output:

```
Shape of first dataset: (11933, 98)
Shape of second dataset: (15560, 161)
Shape after merging: (27493, 188)
Merged file saved as Cleaning_data_17_covid_23.csv
```

Cell 7: ■ Markdown

----- Uptil here-----

Cell 8: ■ Code

```
combined_df['RIDRETH3'].value_counts()
```

Output:

```
RIDRETH3
3.0      11488
4.0       5695
1.0       3107
2.0       2917
6.0       2319
7.0       1967
Name: count, dtype: int64
```

Cell 9: ■ Code

```
# Pie chart for race/ethnicity distribution (RIDRETH3)
import matplotlib.pyplot as plt

# Using combined_df and RIDRETH3 column
```

```

race_counts = combined_df['RIDRETH3'].value_counts().sort_index()

# Map codes to descriptions as per your provided list
race_labels = {
1: 'Mexican American',
2: 'Other Hispanic',
3: 'Non-Hispanic White',
4: 'Non-Hispanic Black',
5: 'Non-Hispanic Asian', # Note: Your list has 6 for Asian, but data uses 5
6: 'Non-Hispanic Asian', # Matching your list
7: 'Other Race - Including Multi-Racial' # As per your list (not "Refused")
}

# Prepare labels and values
labels = [race_labels.get(code, f'Code {code}')] for code in race_counts.index]
sizes = race_counts.values

# Create pie chart
fig, ax = plt.subplots(figsize=(10, 8))
ax.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=90,
colors=plt.cm.tab10.colors)
ax.set_title('Race/Ethnicity Distribution (RIDRETH3)', fontsize=14,
fontweight='bold')
ax.axis('equal') # Equal aspect ratio ensures that pie is drawn as a circle.

plt.tight_layout()
plt.savefig('race_ethnicity_pie_chart.png', dpi=300, bbox_inches='tight')
plt.show()

print("Pie chart saved as 'race_ethnicity_pie_chart.png'")
print(f"Total participants: {race_counts.sum()}")

# Display percentage table
print("\n" + "="*60)
print("RACE/ETHNICITY DISTRIBUTION TABLE")
print("="*60)
print(f"{'Code':<5} {'Description':<35} {'Count':>8} {'Percentage':>10}")
print("-"*60)

```

```

for code in race_counts.index:
    desc = race_labels.get(code, f'Code {code}')
    count = race_counts[code]
    pct = (count / race_counts.sum()) * 100
    print(f"{code:<5} {desc:<35} {count:>8,} {pct:>9.2f}%")
    print("-"*60)

```

Output:

Pie chart saved as 'race_ethnicity_pie_chart.png'

Total participants: 27493

=====

RACE/ETHNICITY DISTRIBUTION TABLE

=====

Code	Description	Count	Percentage
1.0	Mexican American	3,107	11.30%
2.0	Other Hispanic	2,917	10.61%
3.0	Non-Hispanic White	11,488	41.79%
4.0	Non-Hispanic Black	5,695	20.71%
6.0	Non-Hispanic Asian	2,319	8.43%
7.0	Other Race - Including Multi-Racial	1,967	7.15%

=====

[STDERR]

<string>;30: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown

Race/Ethnicity Distribution (RIDRETH3)

